QUESTION ONE: ENERGY CHANGES

A steel ball of mass **0.125 kg** is launched vertically upward from the ground with an initial speed of **18.0 m/s**. Air resistance acts on the ball throughout its motion.

(a)

Analyse the energy changes that occur from launch until the ball reaches its maximum height.

Your answer must include:

- The forms of energy involved
- How energy is transferred
- The effect of air resistance on the energy distribution

(b)

Calculate the maximum height the ball would reach **if air resistance were not present**.

(c)

In reality, the ball reaches a height of only **14.2 m**.

Determine the amount of mechanical energy lost due to air resistance during the upward motion, and explain **where this energy goes**.

(d)

The ball falls back down. At a height of **6.0 m** above the ground, its downward speed is measured to be **12.5 m/s**.

Analyse whether this speed is greater or less than what would be expected without air resistance, and justify your conclusion using energy concepts.

(e)

A student now throws the ball straight upward again, but this time the ball is coated in oil, increasing air resistance.

Explain how this change affects:

- The maximum height
- The time of flight

The distribution of energy transformations

- Your answer must include **why** each change occurs.

QUESTION TWO: THERMAL ENERGY

A metal container holds **1.20 kg** of water initially at **12°C**. A heater supplies a constant **85 W** of thermal power to the water. The specific heat capacity of water is **4200 J kg^{-1} °C^{-1}**.

The container itself absorbs **15%** of the heater's energy.

(a)

Analyse the meaning of *specific heat capacity* and explain how it applies to the heating of the water.

(b)

Calculate the time required for the water to reach **55°C**.

(c)

During heating, the water is stirred continuously.

Explain how stirring affects the heating process, and analyse why the temperature rises more uniformly compared to unstirred water.

(d)

After reaching **55°C**, the heater continues running. Eventually, the water begins to boil. Explain why the temperature remains constant during boiling, even though energy is still being supplied.

Your answer must reference **latent heat** and **energy distribution**.

(e)

The heater is switched off once boiling begins.
The water cools from **100°C** to **70°C** over **25 minutes** in a room at **20°C**.

Analyse the cooling process, explaining:

- Why the cooling rate is fastest at the beginning
- Why the rate slows down

The types of heat transfer involved

- Your answer must reference **temperature difference**, **energy flow**, and **thermal equilibrium**.

QUESTION THREE: ELECTRICITY

A circuit contains a power supply, a fixed resistor, a variable resistor, and a heating coil submerged in oil. The power supply can output up to **12 V**.
The heating coil has a resistance of **8.0 Ω**.

The variable resistor can be adjusted between **0–20 Ω**.

(a)

Draw (describe) where an ammeter and voltmeter should be placed to measure:

- The total circuit current

The voltage across the heating coil

- Explain **why** each meter must be placed in that position.

(b)

The variable resistor is set to **12 Ω**.

Calculate the circuit current and the electrical power delivered to the heating coil.

(c)

The oil surrounding the heating coil warms up.

Analyse why the coil is submerged in oil rather than air, referring to:

- Thermal conductivity
- Heat capacity
- Safety considerations

(d)

The variable resistor is increased gradually from **12 Ω** to **20 Ω**.

Analyse how this change affects:

- The circuit current
- The voltage across the heating coil

The heating rate of the oil

- Your answer must reference **Ohm's law**, **series circuits**, and **energy transfer**.

(e)

The heating coil is replaced with a coil of **higher resistance**.

Explain and analyse how this change affects the circuit behaviour, including:

- Current
- Power output

Heating effectiveness

- Your answer must include **quantitative reasoning** and **energy concepts**.